

Carbohydrates

1. Given below are two statements :

Statement-I: Glycogen is similar to amylose in its structure.

Statement-II: Glycogen is found in yeast and fungi also.

In the light of the above statements, choose the correct answer from the options given below: (2024 Re)

- Statement-I is true but Statement-II is false.
- Statement-I is false but Statement-II is true.
- Both Statement-I and Statement-II are true.
- Both Statement-I and Statement-II are false.

2. The reagents with which glucose does **not** react to give the corresponding tests/products are (2024)

- | | |
|---------------------|---------------------------|
| A. Tollen's reagent | B. Schiff's reagent |
| C. HCN | D. NH_2OH |
| E. NaHSO_3 | |

Choose the correct options from the given below:

- B and E
- E and D
- B and C
- A and D

3. Sucrose on hydrolysis gives: (2020)

- α -D-Glucose + β -D-Glucose
- α -D-Glucose + β -D-Fructose
- α -D-Fructose + β -D-Fructose
- β -D-Glucose + α -D-Fructose

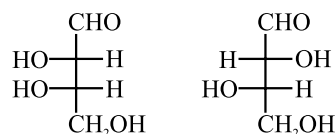
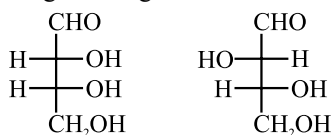
4. Which of the following statement is not true about glucose?

- It contains five hydroxyl groups (2020-Covid)
- It is a reducing sugar
- It is an aldopentose
- It is an aldohexose

5. The difference between amylose and amylopectin is (2018)

- Amylopectin have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ α -linkage
- Amylose have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ β -linkage
- Amylose is made up of glucose and galactose
- Amylopectin have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ β -linkage

6. The correct corresponding order of names of four aldoses with configuration given below: (2016 - II)

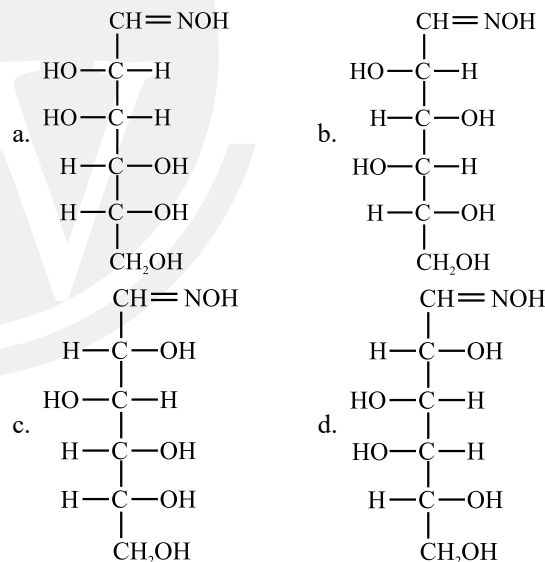


- L-erythrose, L-threose, L-erythrose, D-threose
- D-threose, D-erythrose, L-threose, L-erythrose
- L-erythrose, L-threose, D-erythrose, D-threose
- D-erythrose, D-threose, L-erythrose, L-threose

7. Which one given below is a non-reducing sugar? (2016 - I)

- Sucrose
- Maltose
- Lactose
- Glucose

8. D(+) glucose reacts with hydroxyl amine and yields an oxime. The structure of the oxime would be: (2014)



Proteins

9. The **incorrect** statement about denaturation of proteins is : (2022 Re)

- Uncoiling of the helical structure takes place.
- It results due to change of temperature and/ or pH
- It results in loss of biological activity of proteins.
- A protein is formed from amino acids linked by peptide bonds.

10. Which of the following is a basic amino acid? (2020)

- Alanine
- Tyrosine
- Lysine
- Serine

11. The non-essential amino acid among the following is: (2019)

- a. Valine b. Leucine
c. Alanine d. Lysine

12. Which of the following compounds can form a zwitter ion? (2018)

- a. Aniline b. Acetanilide
c. Glycine d. Benzoic acid

13. In a protein molecule, various amino acids are linked together by: (2016 - I)

- a. Dative bond b. α -glycosidic bond
c. β -glycosidic bond d. Peptide bond

Enzymes and Vitamins

14. Cheilosis occurs due to deficiency of

(2023-Manipur)

- a. thiamine b. nicotinamide
c. pyridoxamine d. riboflavin

15. The **incorrect** statement regarding enzymes is

(2022)

- a. Enzymes are very specific for a particular reaction and substrate
b. Enzymes are biocatalysts
c. Like chemical catalysts enzymes reduce the activation energy of bio process.
d. Enzymes are polysaccharides.

16. The RBC deficiency is deficiency disease of: (2021)

- a. Vitamin B₆ b. Vitamin B₁
c. Vitamin B₂ d. Vitamin B₁₂

17. Deficiency of which vitamin causes osteomalacia? (2020-Covid)

- a. Vitamin D b. Vitamin K
c. Vitamin E d. Vitamin A

Nucleic Acids

18. Given below are two statements: (2023)

Statement-I: A unit formed by the attachment of a base to 1' position of sugar is known as nucleoside

Statement-II: When nucleoside is linked to phosphorous acid at 5'-position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement-I is false but Statement-II is true.
b. Both Statement-I and Statement-II are true.
c. Both Statement-I and Statement-II are false.
d. Statement-I is true but Statement-II is false.

19. The central dogma of molecular genetics states that the genetic information flows from: (2016 - II)

- a. Amino acids $\rightarrow$ Proteins $\rightarrow$ DNA
b. DNA $\rightarrow$ Carbohydrates $\rightarrow$ Proteins
c. DNA $\rightarrow$ RNA $\rightarrow$ Proteins
d. DNA $\rightarrow$ RNA $\rightarrow$ Carbohydrates

20. The correct statement regarding RNA and DNA, respectively is: (2016 - I)

- a. The sugar component in RNA is 2'-deoxyribose and the sugar component in DNA is arabinose
b. The sugar component in RNA is arabinose and the sugar component in DNA is 2'-deoxyribose
c. The sugar component in RNA is ribose and the sugar component in DNA is 2'-deoxyribose
d. The sugar component in RNA is arabinose and the sugar component in DNA is ribose

Hormones

21. Which of the following hormones is produced under the conditions of stress which stimulate glycogenolysis in the liver of human beings? (2014)

- a. Thyroxin b. Insulin c. Adrenaline d. Estradiol

Answer Key

1. (b) 2. (a) 3. (b) 4. (c) 5. (a) 6. (d) 7. (a) 8. (c) 9. (d) 10. (c)
11. (c) 12. (c) 13. (d) 14. (d) 15. (d) 16. (d) 17. (a) 18. (d) 19. (c) 20. (c)
21. (c)